

Application of the Millepede algorithm to the NeuLAND time-position calibration

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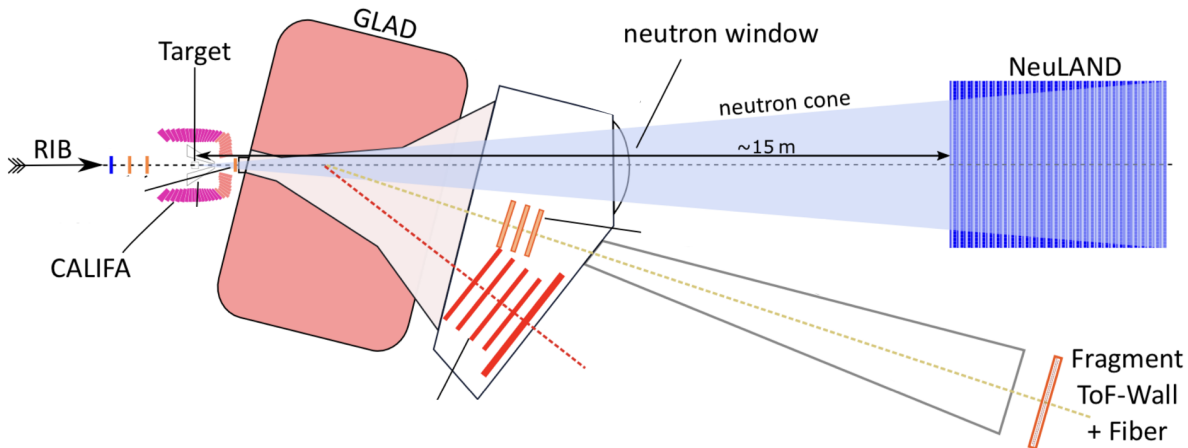
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NeuLAND setup in R³B at FAIR^[1]



^[1] K. Boretzky *et al.*, Nucl. Instrum. Methods Phys. Res., Sect. A **1014**, 165701 (2021).

NeuLAND setup in R³B at FAIR^[1]

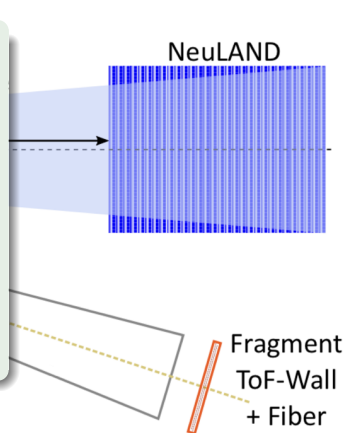


Geometry:

- 26 planes
- $250 \times 250 \text{ cm}^2$
- 50 scintillators each plane
- 1300 Bars in total
- 2600 PMTs in total

Output:

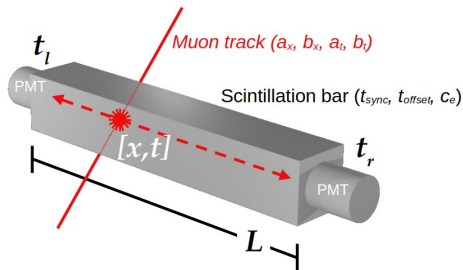
- Time values
- Positions



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NeuLAND time and position calibration

Calibration with cosmic muons:



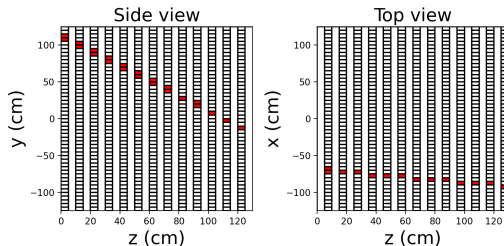
3900 parameters from scintillators and $6N$ parameters from N muon tracks

Calibration equations

$$b_x + z \cdot a_x + c_e \cdot t_{\text{diff}}/2 - t_{\text{offset}}/2 = 0 \quad (1)$$

$$b_t + z \cdot a_t + t_{\text{sync}} + \frac{\mathcal{L}}{2 \cdot c_e} - t_{\text{sum}}/2 = 0 \quad (2)$$

Current method: using **active bar positions** to reconstruct muon tracks:



Equations with known muon tracks

$$\mathcal{C}(a_x, b_x, z) + c_e \cdot t_{\text{diff}} - t_{\text{offset}} = 0 \quad (3)$$

$$\mathcal{C}(a_t, b_t, z, t_{\text{sum}}) \cdot c_e + 2 \cdot t_{\text{sync}} \cdot c_e + \mathcal{L} = 0 \quad (4)$$

Objective function minimization

$$\vec{P}, \vec{Q} = \underset{\vec{p}, \vec{q}}{\operatorname{argmin}} f(\vec{p}, \vec{q}, z, t_{\text{diff}}, t_{\text{sum}})$$

where

$\vec{p}$: Parameters of scintillators

$\vec{q}$: Parameters of all muon tracks ($\vec{q}^0, \vec{q}^1, \dots, \vec{q}^N$)

Minimization with Newton's method

Objective function minimization

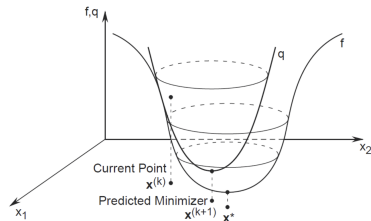
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Newton's method



Limitations of Newton's method

- Evaluation of Hessian is very expensive. (Quasi Newton's methods, e.g. BFGS)
- Expensive computation with a very large number of parameters.

Dimension of $\mathcal{B}(\vec{w})$ when using N muon tracks:

$$3900 + 6N \sim \mathcal{O}(N)$$

n^{th} iteration for all parameters

$$\vec{W}_{n+1} = \vec{W}_n - \eta \cdot \mathcal{B}(\vec{w})^{-1} \nabla f(\vec{w}) \Big|_{\vec{w}=\vec{W}_n}$$

where

$$\vec{w} = (\vec{p}, \vec{q}^0, \vec{q}^1, \dots, \vec{q}^N)$$

$$\eta = 1 \quad \text{(Line search)}$$

$$\mathcal{B}(\vec{w}) = \nabla^2 f(\vec{w}) \quad \text{(Hessian matrix)}$$

Special features in calibration with muon tracks

- Only **3900** parameters from scintillation bars are needed.
- Parameters related to one muon track is independent to another.

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Separation into **global** and **local** parameters in Millepede algorithm:

Global parameters $\vec{p}_i$	Local parameters $\vec{q}^0$	...	Local parameters $\vec{q}^N$
$p_1, p_2, \dots, p_{3900}$	$a_x^0, b_x^0, a_y^0, b_y^0, a_t^0, b_t^0$	...	$a_x^N, b_x^N, a_y^N, b_y^N, a_t^N, b_t^N$

Minimization with Millepede algorithm

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n^{th} iteration for global parameters^[1]

$$\vec{p}_{n+1} = \vec{p}_n - \mathcal{B}(\vec{p}, \vec{q})^{-1} (\nabla_p f + \mathcal{R}(\vec{p}, \vec{q}) \nabla_q f) \Big|_{\vec{p}=\vec{p}_n, \vec{q}=\vec{q}^n}$$

where:

$$\mathcal{B}(\vec{p}, \vec{q}) = \nabla_p^2 f - \nabla_{pq}^2 f \cdot \nabla_q^2 f \cdot \left(\nabla_{pq}^2 f \right)^T \quad (\text{size: } 3900 \times 3900)$$

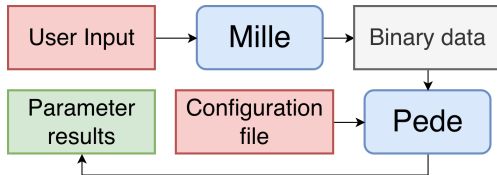
$$\mathcal{R}(\vec{p}, \vec{q}) = -\nabla_{pq}^2 f \cdot \left(\nabla_q^2 f \right)^{-1} \quad (\text{size: } 3900 \times 6)$$

Advantages of Millepede algorithm

- Still simultaneous fitting of all parameters
- Computation complexity independent of the number of muon tracks

^[1] For the full derivation, please visit Millepede-II manual (url: https://www.desy.de/~kleinwrt/MP2/doc/html/draftman_page.html)

Millepede-II program structure



Prerequisites

- Calibration equation must be **linear**.

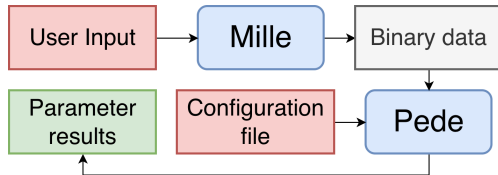
$$M(z, t) = \sum_i G^i(z, t) p^i + \sum_j L^j(z, t) q^j$$

- "Good" initial global parameter values.
- No rank deficit.

^[1] Full documentation can be seen via
https://yanzhaow.github.io/R3BRoot/neuland_cal.html

Millepede-II program and adaptation to NeuLAND^[1]

Millepede-II program structure



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Rank deficit in calibration equations

```
Chi^2/Ndf = 2752.53 (should be close to 1)
=> multiply all input standard deviations by factor 52.46

Rank defect = 1 for global matrix, should be 0
=> please provide correct mille data

WarningWarningWarningWarningWarningWarningWarningWarningWarn
arningWarningWarningWarningWarningWarningWarningWarningWarn
```

A simple example:

$$y = a \cdot x + b + c$$

With three data points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) :

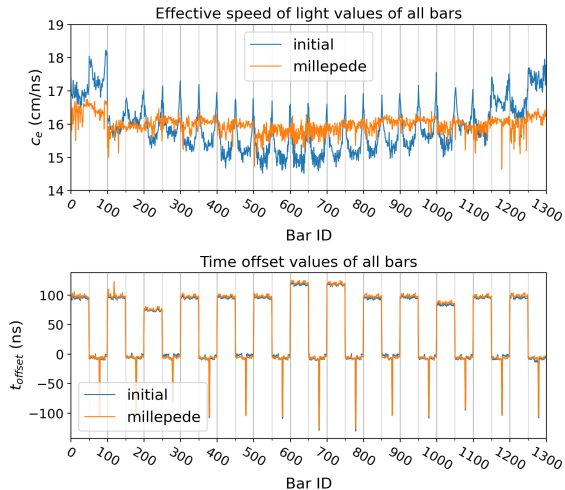
$$\begin{bmatrix} x_1 & 1 & 1 \\ x_2 & 1 & 1 \\ x_3 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Solution: Fixing either parameter b or c .

Solving the rank deficit in NeuLAND calibration

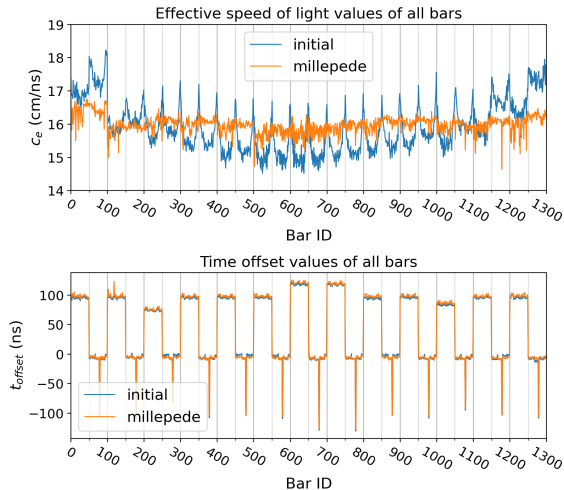
- Assuming muon speed to be **c**. (Fixing parameter a_t)
- Time synchronization factor fixed to be 0 for one bar.

Comparison between initial values and results

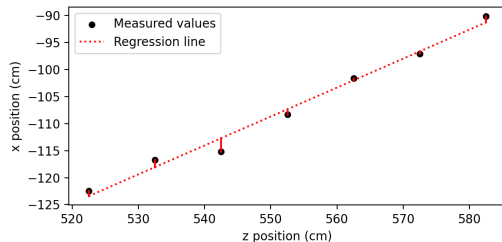


Calibration results from Millepede

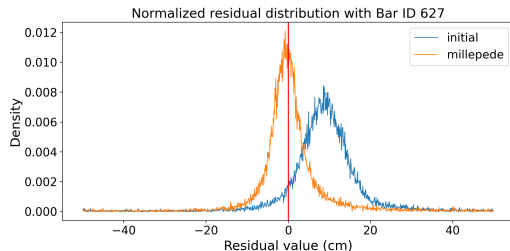
Comparison between initial values and results



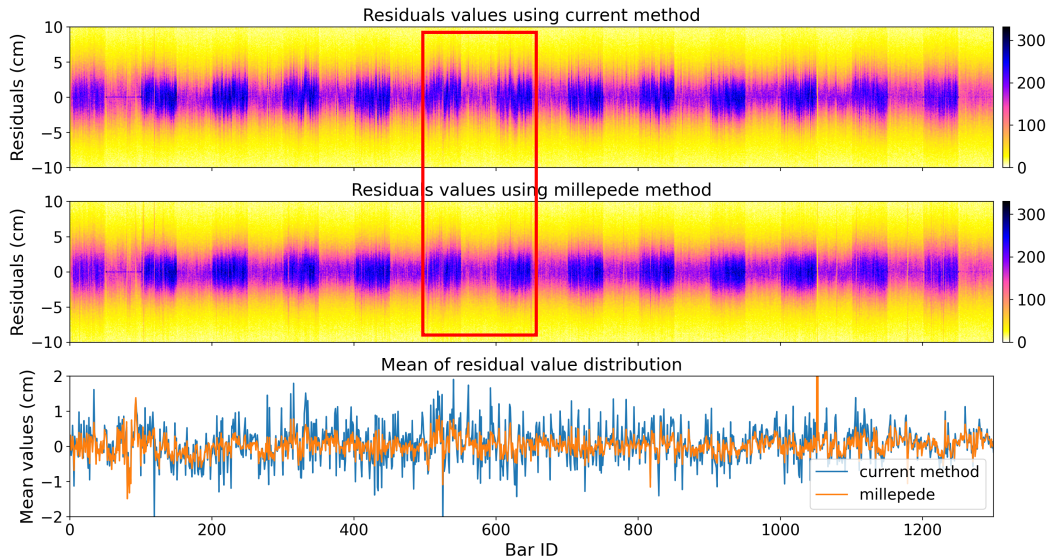
Verification through reconstructed muon track



Residual = predicted value – measured value



Comparison with the current method



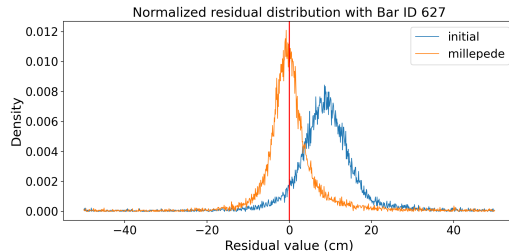
Summary and outlook

Summary

- Millepede algorithm reduces the computation complexity of the calibration processes using cosmic muons.
- New method using Millepede algorithm produce less biased results compared to the current method.

Outlook

- Implementing the millepede algorithm with modern C++ for a better integration with the current software framework.
- Applying millepede algorithm to the energy related parameters as well.



[centipede.cpp](https://github.com/YanzhaoW/centipede.cpp) Public

A C++ implementation of the Millepede method.

experiment

physics

calibration

data-analysis

cpp23

C++

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MIT License

Updated on Feb 8

URL: <https://github.com/YanzhaoW/centipede.cpp>